

Task:

Communication

Protocol AXI-Lite

Subordinate

Insert Name, OU Unit,
Date

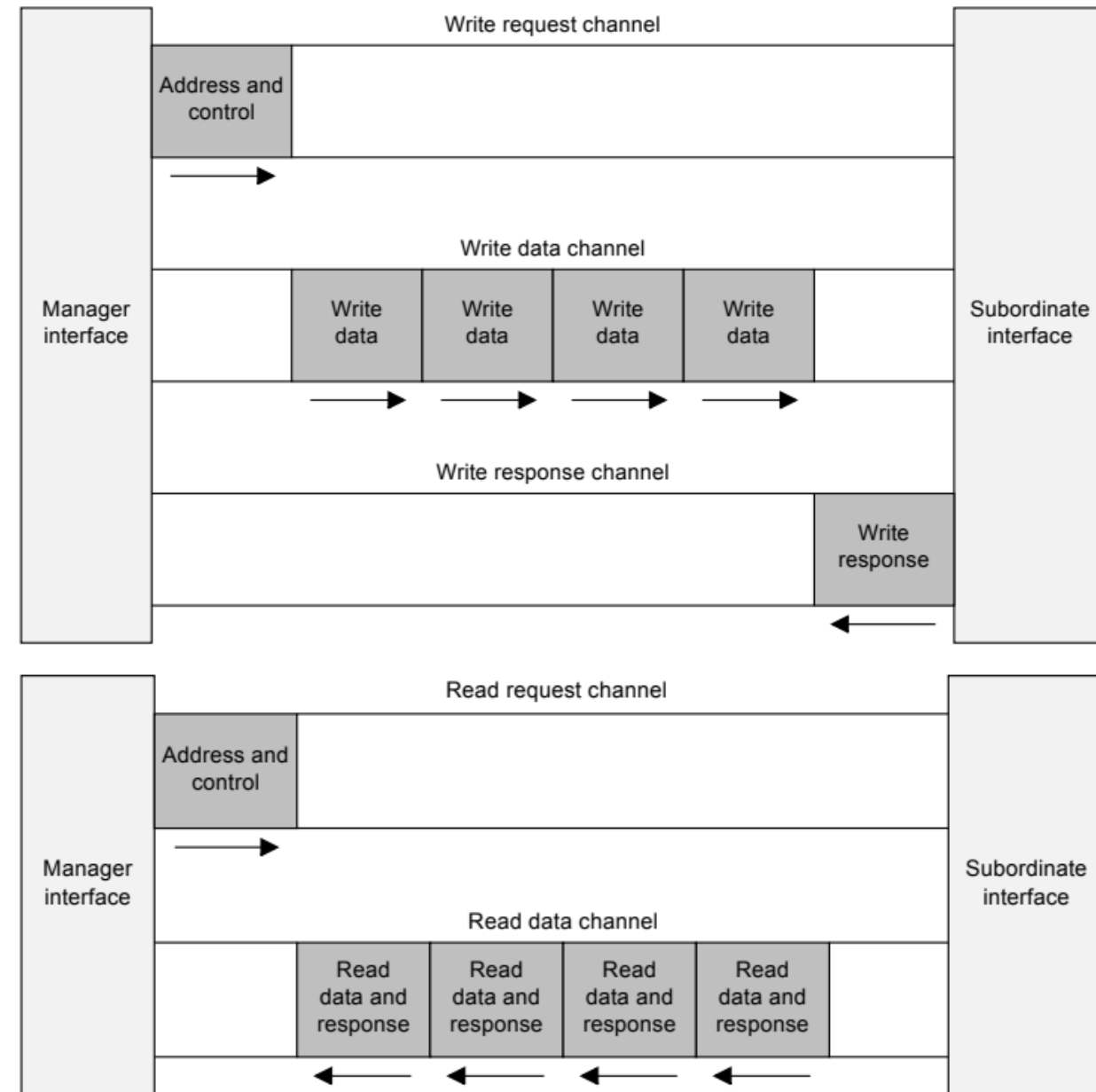
- 1. AXI Protocol**
- 2. Task AXI-Lite Subordinate**

Introduction to the AXI protocol

- Developed by ARM and part of the AMBA specification
- On chip bus communication protocol
- Communication of the AXI bus is between a single Manager and Subordinate but it is also possible to have a interconnect to connect multiple Managers and Subordinates together
- Key features of the AXI protocol:
 - Suitable for high-bandwidth and low latency
 - Separate address and data phases
 - Separate write and read channels
 - Support of issuing multiple address

Architecture of AXI

- Interface between a Manager and Subordinate
- AXI has five independent channels
 - Write reques (signals starting with AW)
 - Write data (signals startitng with W)
 - Write response(signals starting with B)
 - Read request (signals starting with AR)
 - Read data (signals starting with R)
- Data is transferred either by a write or read request
- Each channel uses a handshake protocol for the data transfer
- AXI allows for transfer of multiple data for one transfer request

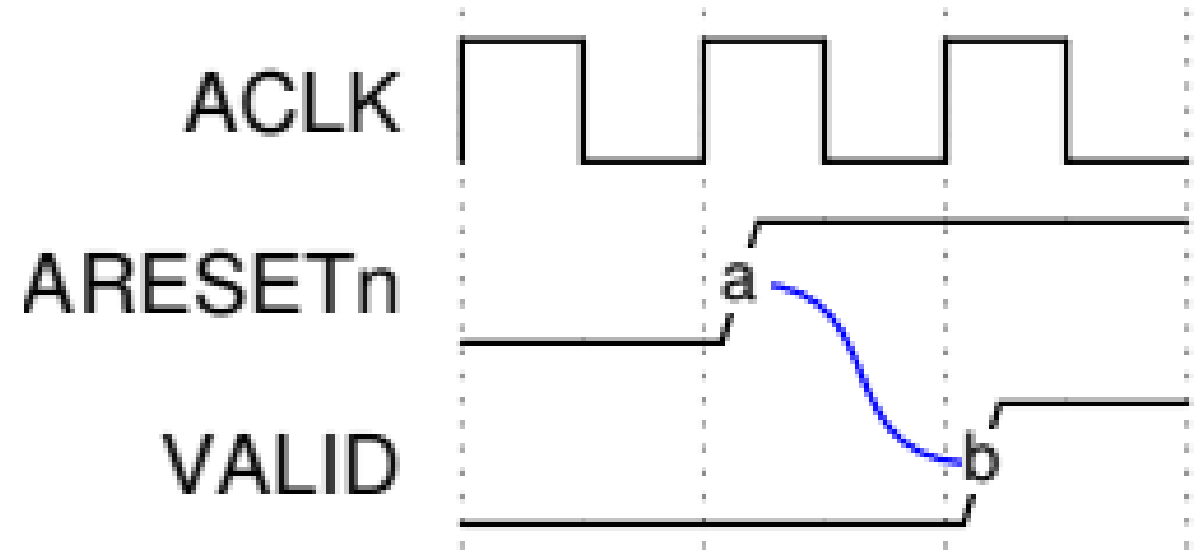


Difference between AXI and AXI-Lite

- The main difference between AXI and AXI-Lite is that AXI supports burst transfer while AXI-Lite doesn't
- This for each AXI-Lite write or read transfer only one data word is transmitted
- => AXI-Lite is a simplified version of AXI
- AXI-Lite doesn't support every feature like burst transfer
=> Easier to implement and uses less resources

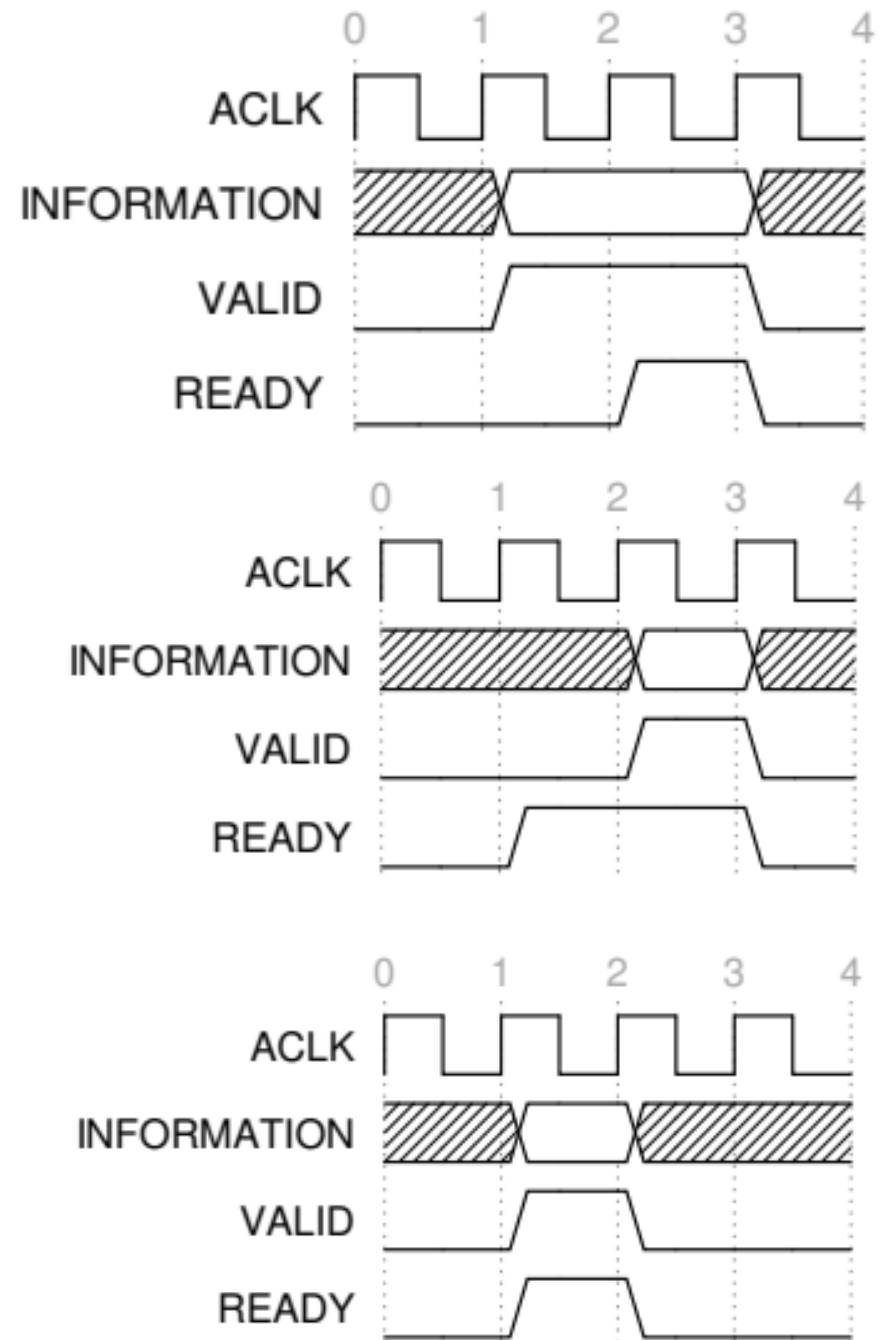
Clock and reset

- AXI clock
 - Each AXI interface has a single clock signal ACLK
 - All input signals are sampled on the rising edge
 - All output signal can only change after the rising edge
- **Important** no combinatorial paths between Manager and Subordinate (allows for adding register to help with timing)
- AXI reset
 - Active LOW reset signal ARESETn
 - Reset signal can be asynchronously
 - Signal deassertation can only be synchronous with a rising edge of the clock
 - AWVALID, WVALID, ARVALID, BVALID, RVALID signals should be driven low



AXI Handshake

- All channels use the same VALID/READY handshake process for data/information transfer
- Both Manager and Subordinate can control the rate of the information movement between them
- The source of the data/information asserts the VALID signal when there is valid data/information on the channels signals
- In AXI the source module isn't allowed to deassert the VALID signal once it was asserted
- The destination asserts the READY signal when it can accept the information
- Transfer occurs when both VALID and READY are HIGH on the rising edge of the clock



Write Address Channel

- Used to request a Write transaction and send the required information to the subordinate
- The information about the address of the transfer is given and the ID number of the transfer
- For AXI also the information about the length of the transfer, the kind of Burst transfer and how many bytes are transfer for each transfer is transmitted
- The important signals of this channel are AWADDR, AWSIZE, AWBURST, AWLENGTH, AWID
- For AXI-Lite only AWADDR and AWID are mandatory and the handshake signals

Signal Name	Width
AWADDR	Address width
AWVALID	1
AWREADY	1
AWID	>0

Write Data channel

- The write Data channel is used to transfer the data that has to be written in the given address
- The important signals are WDATA andWSTRB for AXI-Lite
- For AXI the WLAST signal is also mandatory
- TheWSTRB signal indicate which bytes from WDATA have valid information

Signal Name	Width
WDATA	Data width
WVALID	1
WREADY	1
WSTRB	Data width /8
WID	>0

Write Response Channel

- In the write response channel, the subordinates gives information about the success of the write transfer
- The default value is 0 which indicates a successful write transfer
- The BRESP signal can be larger than 1 bit if more error codes are needed
- The important signals of this channel are the handshake signals BVALID and BREADY and the BRESP signal

Signal Name	Width
BRESP	1
BVALID	1
BREADY	1
BID	>0

Read Address channel

- The read address channel works the same as the write address channel
- The only difference is that it request a read transfer from the subordinate and the signal names are changed

Signal Name	Width
ARADDR	Address width
ARVALID	1
ARREADY	1
ARID	>0

Read Data Channel

- The Read data channels differs from the write data channel
- Here the subordinate is the provider of the information and the read response is also provided in this channel instead of a sperate channel
- Like for the write response channel a b000 on RRESP indicates a correct data transfer

Signal Name	Width
RDATA	Data width
RVALID	1
RREADY	1
RRESP	3
RID	>0

Task

- Your task is to write an AXI-Lite subordinate for this a Verilog file is provided and you must write the logic that handles the AXI-Lite protocol
- There is also a Testbench provided for testing your module
- The subordinate should at least have one register you can write to and read from using the AXI-Lite protocol
- For more Information about the working function of the AXI protocol you can look at the AXI specification document or other online resources
- If you want, you can also implement the strobe function of AXI, but it is not necessary
- Tips:
 - Use Case statement for implementing the address/register writing reading
 - Have signals to check if read/write transfer has started and when it is finished

